

HMIC 2018

April 09, 2018

HMIC 2018

1. [6] Let $m > 1$ be a fixed positive integer. For a nonempty string of base-ten digits S , let $c(S)$ be the number of ways to split S into contiguous nonempty strings of digits such that the base-ten number represented by each string is divisible by m . These strings are allowed to have leading zeroes.

In terms of m , what are the possible values that $c(S)$ can take?

For example, if $m = 2$, then $c(1234) = 2$ as the splits 1234 and $12|34$ are valid, while the other six splits are invalid.

Proposed by: Kevin Sun

Answer: 0 and 2^n for all nonnegative integer n

First, we note that $c(1) = 0$ and $c(00\dots 0) = 2^{n-1}$ if there are n zeroes in the string. Now we show that these are the only possibilities. Note that a split can be added if and only if the string before this split (ignoring all other splits) represent a multiple of m (if there is a split before it, then removing the digits before this preceding split is equivalent to subtracting the removed number times a power of 10, which will also be a multiple of m , so the remaining number between the two splits remain a multiple of m). Thus, whether we can add a split or not depends only on the string itself and no other splits, so $c(S)$ is 0 if the number is not divisible by m and a power of two otherwise.

2. [7] Consider a finite set of points $T \in \mathbb{R}^n$ contained in the n -dimensional unit ball centered at the origin, and let X be the convex hull of T . Prove that for all positive integers k and all points $x \in X$, there exist points $t_1, t_2, \dots, t_k \in T$, not necessarily distinct, such that their centroid

$$\frac{t_1 + t_2 + \dots + t_k}{k}$$

has Euclidean distance at most $\frac{1}{\sqrt{k}}$ from x .

(The n -dimensional unit ball centered at the origin is the set of points in $\mathbb{R}^n$ with Euclidean distance at most 1 from the origin. The convex hull of a set of points $T \in \mathbb{R}^n$ is the smallest set of points X containing T such that each line segment between two points in X lies completely inside X .)

Proposed by: Henrik Boecken

By the definition of convex hull, we can write $x = \sum_{i=1}^m \lambda_i z_i$, where each $z_i \in T$, each $\lambda_i \geq 0$ and $\sum_{i=1}^m \lambda_i = 1$. Consider then a random variable Z that takes on value z_i with probability λ_i . We have $\mathbb{E}[Z] = x$. Let $\bar{Z} = \frac{1}{k} \sum_{i=1}^k Z_i$, where each Z_i is an independent copy of Z . Then we wish to compute

$$\text{Var}[\bar{Z}] = \frac{1}{k^2} \sum_{i=1}^k \text{Var}[Z_i]$$

Finally, we have

$$\text{Var}[Z_i] = \mathbb{E}[|Z_i - x|^2] = \mathbb{E}[|Z_i|^2] - x^2 \leq \mathbb{E}[|Z_i|^2] \leq 1$$

The second equality follows from the identity $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. Now, we know that

$$\mathbb{E}[|x - \frac{1}{k} \sum_{i=1}^k Z_i|^2] = \text{Var}[\bar{Z}] \leq \frac{1}{k}$$

Thus, there must exist some realization of x_i of the Z_i such that

$$|x - \frac{1}{k} \sum_{i=1}^k x_i|^2 \leq \frac{1}{k}$$

and we are done.

3. [8] A polygon in the plane (with no self-intersections) is called *equitable* if every line passing through the origin divides the polygon into two (possibly disconnected) regions of equal area.

Does there exist an equitable polygon which is not centrally symmetric about the origin?

(A polygon is centrally symmetric about the origin if a 180-degree rotation about the origin sends the polygon to itself.)

Proposed by: Kevin Sun

Consider the polygon with vertices

$$\begin{aligned} &A(1, 0), B(0, 1), C(0, 5), D(5, 0), E(7, 0), \\ &F(0, -7), G(-7, 0), H(-\sqrt{84}, 0), I(0, \sqrt{84}), J(0, 6), \\ &K(-6, 0), L(-5, 0), M(0, -5), N(0, -1). \end{aligned}$$

Notice that its intersection with each of the four quadrants are all trapezoids with angles $45^\circ, 45^\circ, 135^\circ, 135^\circ$, and the trapezoids in quadrant 1 and 3 both have area $\frac{5^2-1^2}{2} = \frac{7^2-5^2}{2} = 12$, and the trapezoids in quadrant 2 and 4 both have area $\frac{7^2-1^2}{2} = \frac{84-6^2}{2} = 24$, so by similar triangles, we can show that this polygon is indeed equitable. It is also apparent that this polygon is not centrally symmetric.

4. [10] Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x + f(y + xy)) = (y + 1)f(x + 1) - 1$$

for all $x, y \in \mathbb{R}^+$.

($\mathbb{R}^+$ denotes the set of positive real numbers.)

Proposed by: Ashwin Sah

Let $P(x, y)$ denote the assertion that

$$f(x + f(y + xy)) = (y + 1)f(x + 1) - 1.$$

Claim 1. f is injective. Proof: If $f(a) = f(b)$ then $P\left(x, \frac{a}{x+1}\right), P\left(x, \frac{b}{x+1}\right)$ yields $a = b$, since $f(x + 1) \in \mathbb{R}^+$ so in particular is nonzero. $\circ$ Now $P\left(x, \frac{1}{f(x+1)}\right)$ yields

$$f\left(x + f\left(\frac{x+1}{f(x+1)}\right)\right) = f(x+1)$$

hence by injectivity

$$x + f\left(\frac{x+1}{f(x+1)}\right) = x + 1$$

so that

$$f\left(\frac{x+1}{f(x+1)}\right) = 1.$$

By injectivity, this equals some constant c , so that

$$\frac{x+1}{f(x+1)} = c$$

for all $x \in \mathbb{R}^+$. Now letting $x, y > 1$ in $P(x, y)$ automatically yields

$$\frac{x}{c} + \frac{y+xy}{c^2} = (y+1)\left(\frac{x+1}{c}\right) - 1,$$

which immediately yields $c = 1$ if we take x, y large. Finally, we have $f(x+1) = x+1 \forall x \in \mathbb{R}^+$. Finally, $P\left(x, \frac{y}{x+1}\right)$ yields

$$f(x + f(y)) = \left(\frac{y}{x+1} + 1\right)f(x+1) - 1 = x + y,$$

so that fixing y and letting $x > 1$ yields

$$x + f(y) = x + y,$$

so that $f(y) = y$. This was for arbitrary positive y , so that $f(x) = x \forall x \in \mathbb{R}^+$, which clearly works.

5. [11] Let G be an undirected simple graph. Let $f(G)$ be the number of ways to orient all of the edges of G in one of the two possible directions so that the resulting directed graph has no directed cycles. Show that $f(G)$ is a multiple of 3 if and only if G has a cycle of odd length.

Proposed by: Yang Liu

Let $f_G(q)$ be the number of ways to color G with q colors. This is the chromatic polynomial of G , and turns out to be polynomial in q . Indeed, choose an edge e and let $G \setminus e$ be the graph G with e removed and let G/e be graph G with the vertices on either side of e merged together (multiple edges are removed). It is not hard to see that

$$f_G(q) = f_{G \setminus e}(q) - f_{G/e}(q)$$

and that for an empty graph E_n with n vertices, we have $f_{E_n}(q) = q^n$. Induction finishes.

Now we can also show that the desired number of ways to direct G to avoid directed cycles is $c_G = (-1)^n f_G(-1)$ where n is the number of vertices of G . The way to do it is to show that

$$c_G = c_{G \setminus e} + c_{G/e},$$

and $c_{E_n} = 1$. Thus $(-1)^n c_G$ will satisfy the desired recurrence, after multiplying the above by $(-1)^n$, and satisfies the correct initial conditions.

Now all we need to know is that $f_G(q)$ has integer coefficients, so that $(-1)^n c_G \equiv f_G(-1) \equiv f_G(2) \pmod{3}$, and $f_G(2) = 0$ when G is not bipartite and is $2^{c(G)}$ where $c(G)$ is the number of connected components of G otherwise. The result follows.